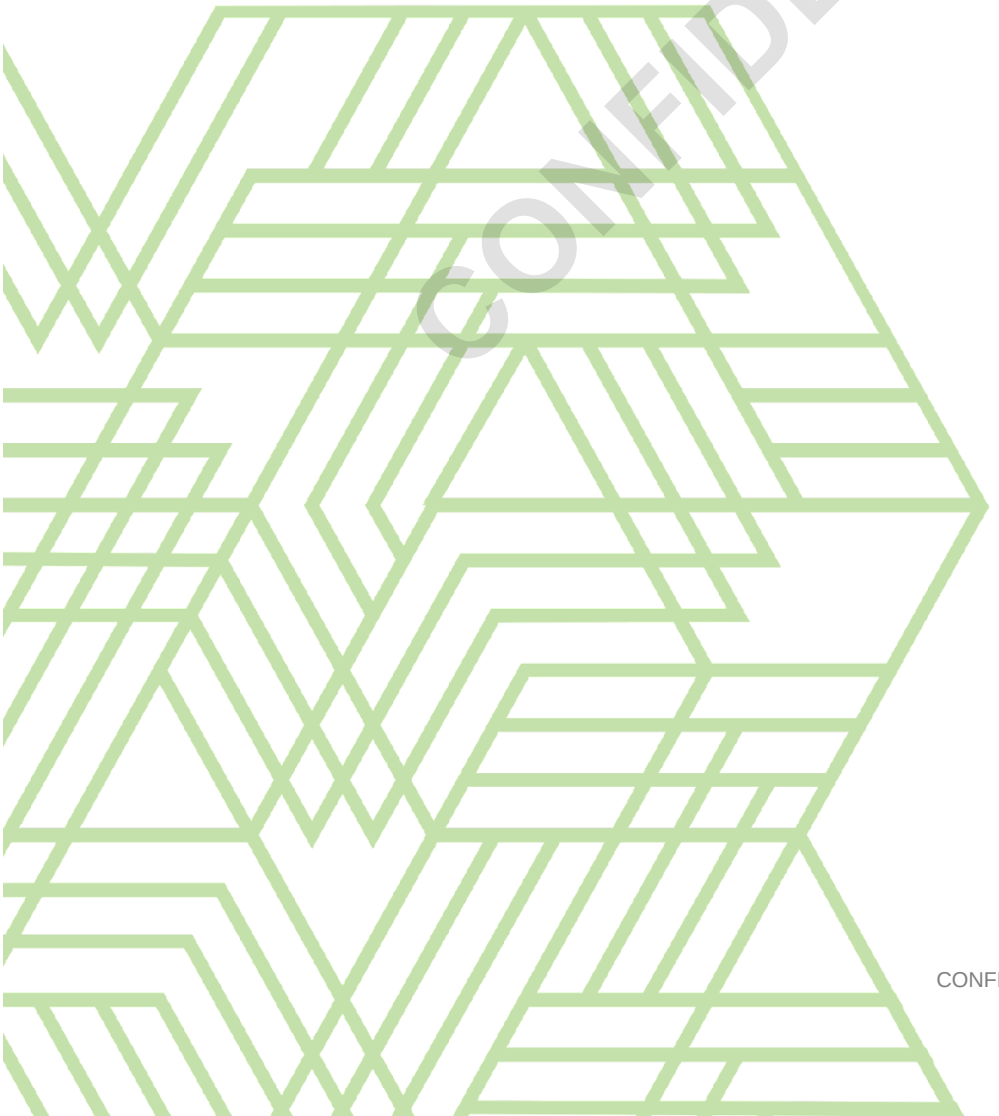
Atmosic SDK 5.4.0.1

Release Notes

SUMMARY: This document provides the release notes for Atmosic SDK 5.4.0.1 and associated tools and utilities enabling design and development of solutions based on ATM2, ATM3, or ATM33/e Wireless SoC Series. It lists the new features and known issues for this SDK release.

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Acronyms and Abbreviations

Acronyms	Definition
API	Application Interface
ATM2	ATM2201 ATM2202 ATM2221 ATM2231 ATM2251
ATM3	ATM3201 ATM3202 ATM3221 ATM3231
ATM33	ATM3330 ATM3325
ATM33e	ATM3330e
ATM33/e	ATM33/ATM33e
ATMWSTK	ATM Wireless Stack
ATT	Attribute Protocol
CTE	Constant Tone Extension
DTM	Direct Test Mode
EATT	Enhanced Attribute Protocol
EVK	Evaluation Kit
FW	Firmware
GATT	Generic Attribute Profile
GAP	Generic Access Profile
HCI	Host Controller Interface
IDE	Integrated Development Environment
MSAP	Monolithic Stack & App Package
NSPE	Non-Secure Processing Environment
OTA	Over-The-Air
OS	Operating System

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Acronyms	Definition
QSPI	Quad Serial Peripheral Interface
RCU	Remote Control Unit
RF	Radio Frequency
SDK	Software Development Kit
SMP:	Secure Manager Protocol
SPE	Secure Processing Environment
SPI	Serial Peripheral Interface

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1. Purpose

This release note provides details on new features/fixes/known issues as part of the SDK 5.4.0.1. Please uninstall any instances of SDK 5.4.0 as this supersedes that version.

Please refer to the README for installation instructions.

This Atmosic SDK 5.4.0.1 supports the following EVK platforms:

EVK	Kit Part Number
Evaluation Kit for ATM2202	ATMEVK-M2202-02
Evaluation Kit for ATM2221	ATMEVK-M2221-02
Evaluation Kit for ATM2251	ATMEVK-M2251-01
Evaluation Kit for ATM3201	ATMEVK-M3201-02
Evaluation Kit for ATM3202	ATMEVK-M3202-02
Evaluation Kit for ATM3221	ATMEVK-M3221-02
Evaluation Kit for ATM3325	ATMEVK-3325-QK
Evaluation Kit for ATM3325 with Extended Storage	ATMEVK-3325-LQK
Evaluation Kit for ATM3330	ATMEVK-3330-QN
Evaluation Kit for ATM3330e	ATMEVK-3330e-QN

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1.1 Development Platforms, IDE and Tool Chains

This section listed the OS and debug tools supported by the ATM33/e series:

Category	Features Supported
Development Platforms	MacOS (x86, M1)
	Linux, Windows 10, Windows 11
IDE	Keil with ARM C++ compiler
	IAR Workbench

Category	Features Supported
	SEGGER Embedded Studio
Debug Tools	J-Link Debugger (on-board for ATM33/e EVKs)
	OpenOCD debug
FW Update	OTA FW Update

Please visit Atmosic Support website for the following documents:

Title	Document Number
ATM33/e Series Evaluation Kit User Guide	ATM33_e-UGEVK
ATM33/e (Rev. B) Series Release Notes	ATM33_e-RNHWB
EVK User Guide for ATM2/ATM3	ATM2_ATM3-UGEVK
ATM33/e Series Register Specification	ATM33_e-RS
Register Specification for ATMx2xx-x1x	ATMx2xx-x1x-RS
ATM33/e Digital Biquad Filter Software Implementation Application Note	ATM33_e-ANDBF
Complete Doxygen-style reference of all APIs provided by this SDK	N/A
Bluetooth LE functionality, including GAP, GATT, and various profiles	N/A

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Additional references for developing applications using third party IDE Software:

Title	Document Number
IAR Workbench User Guide	ATM-UGIAR
Keil MDK Configuration Guide	ATM-CGKMDK
SEGGER Embedded Studio User Guide	ATM-UGSEG
IDE Auxiliary Flash Programming Tool User Guide	ATM-UGIDEFPT

Please submit a support request for access credentials to the Atmosic Support website.

2. New Features / Functionality

2.1 SDK and Application Version

SDK and application versions are displayed as part of the boot up messages:

@009404ab SDK Version: 5.4.0.1

@00940504 APP Version: 0.0.0.9

- SDK version is located in version.h folder in <ROOT>/AtmosicSDK
- App version can be specified using `APP_VERSION` defined during make

2.2 Supported Devices

This release combines the support for all the Atmosic SoC Series - ATM2, ATM3, and ATM33/e.

ATM2		ATM3		ATM33/e
ATM2201-x0x	ATM2201-x1x	ATM3201-x0x	ATM3201-x1x	ATM3330e-5DCAQN
ATM2202-x0x	ATM2202-x1x	ATM3202-x0x	ATM3202-x1x	ATM3330-5DCAQN
ATM2221-x0x	ATM2221-x1x	ATM3221-x0x	ATM3221-x1x	ATM3325-5DCAQK
	ATM2231-x1x		ATM3231-x1x	ATM3325-5LCAQK
	ATM2251-x1x			

2.3 New Features Included in SDK Release 5.4.0.1

This section describes the newly added features supported by SDK release 5.4.0.1 on Rev B silicon

Category	New Features Supported in SDK 5.4.0.1
AoA (BT5.1) Direction Finding	AoA Locator/Receiver

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Category	New Features Supported in SDK 5.4.0.1
Sensor Hub	DataStore, DataTrigger
Clocks	48 MHz
Peripherals	External flash support - QSPI and single-SPI devices (For detailed list of supported devices, please refer to ATM33/e Series Hardware Design GuideList the device supported) Data Storage & OTA Staging
Harvesting Recharge (applicable to ATM3330e only)	Clock control for External Booster Circuit
Energy Harvesting (applicable to ATM3330e only)	Harvesting Power Meter APIs

2.4 ATM33/e

This section describes the complete list of features for the ATM33/e series supported by SDK release 5.4.0.1

Category	Features Supported
AoA/AoD of Arrival (BT5.1)	AoA Tag/Transmitter and AoA Locator/Receiver
Bluetooth LE Core Features	Bluetooth LE Power Control
	Peripheral and Broadcaster role
	Central role (multiple master)
Bluetooth LE 5.2 Features	Enhanced Attribute Protocol (EATT)
Sensor Hub	DataStore, DataTrigger
Quuppa RTLS	Quuppa Tag support
PHY Rates	1 Mbps
	2 Mbps

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Category	Features Supported
	125 kbps / 500 kbps Coded PHY
Clocks	16 MHz
	32 MHz
	48 MHz
	64 MHz
Internal RC	32 kHz Internal RC oscillator (RCOSC)
Audio Interfaces	PDM
Peripherals	I2C Master
	I2S Master
	UART
	SPI Master
	PWM
	IR Tx Support (PWM FIFO based)
	GADC
	QSPI external flash support - data storage, OTA staging
Audio Codecs and PTT/HTT Software support	ADPCM
	Google 1.0 Voice Support
	mSBC (APIs)
	Programmable Filters/Equalizer, Bi-quad support
Power Management Unit	3 V Battery Extension mode (3 V Battery and Supercap)

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Category	Features Supported
	Storage Configuration/Type
	VBAT_LI LDO Enablement: Full operating range; Auto control
Harvesting Recharge (applicable to ATM3330e only)	Direct recharging from harvesting when VBATLI < 3.4 V
	Clock control for External Booster Circuit
	HSC operation/recharge using discrete booster
	LiFePO4 battery 2.5 to 3.4 V (operation+recharging)
	Li-Ion Battery 4.2 V (operation+recharging) EVK default configuration is using VBAT. For Li-Ion (VBATLI) configuration, please refer to ATM33/e Series Evaluation Kit User Guide.
Energy Harvesting (applicable to ATM3330e only)	Standalone Multi-cell PV support
	Standalone Single-cell PV support - 500 mV
	RF - 2.4 GHz
	RF - 914 MHz
	Non-concurrent harvesting (PV or RF)
	Concurrent harvesting without MPPT (multi-cell PV and RF)
	PV/RF charging indicator (charging/not-charging)
	Overvoltage protection of Battery/HSC
	Undervoltage/brownout protection of Battery/HSC
	Harvesting Power Meter
Security	AES-128/256 HW cryptographic engine
	SHA2-HMAC-256 HW cryptographic engine

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Category	Features Supported
	TRNG
	ECDSA for signing (based on MCU boot)
	Secure Debug (driver to lock down debug ports)
	ARM TrustZone
	Root-of-Trust
	Secure Watchdog
	Secure Boot (ECDSA signing/verification)
	Secure firmware upgrade with Anti-rollback

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3. Major Changes / Bug Fixes

Major changes and bug fixes in SDK 5.4.0.1 include:

3.1 ATM2/ATM3/ATM33

3.1.1 ATM Log Level

To provide more flexibility for users to configure the log level for different modules, ATM_LOG_LVL_XXX options have been added to the modules. Each module has its own default log level, which can be modified by adding -DATM_LOG_LVL_(module name)=(level) to the CFLAGS in the makefile.

Size log levels are supported as described below:

Off (O) - No Debug information

Error (E) - Errors only

Warning (W) - Warnings and Errors

Notify (N) - Notifications

Debug (D) - Add description here

Verbose (V) - Verbose

To set (log level), use the first letter of the target level (such as O, E, W, N, D, V). Logs with a higher level (V > D > N > W > E > O), for example the module level of atm_adv.c is set to W, the ATM_LOG print with level V, D, N will be removed when compiling atm_adv.cthan the current module log level will not be compiled into the firmware. An ATM log user can decide the log levels of the modules based on code size requirement and debug information.

- For example, to set the log level of atm_adv.c to warning and debug log level for atm_gap.c, use CFLAGS += -DATM_LOG_LVL_ATM_ADV=W -DATM_LOG_LVL_ATM_GAP=D

3.1.2 DTM HCI 2-Wire Mode

Current HCI and HCI_vendor examples can support DTM HCI 2-wire mode that disables UART flow control. Refer to the README file of that example about this compile option. Note that 2-wire mode still uses the HCI protocol in a limited capacity. This is not the same as the 2-wire protocol (2-byte messages) listed in the Bluetooth LE specification Volume 6, Parf F.

3.1.3 Add Tx Power and Infinite Tx/Rx API

The atm_ble module adds below three new APIs for this purpose when performing RF DTM test.

atm_ble_set_rftest_txpwr

atm_ble_set_rftest_tx_infinite

atm_ble_set_rftest_rx_infinite

3.2 ATM2/ATM3

3.2.1 BOARD Name Setting Change

Change BOARD name setting from mxxxx to ATMEVK_Mxxxx.

Example: change **`make BOARD=m2201`** to **`make BOARD=ATMEVK_M2201`**

***Note**: Only applicable to ATM22xx-x1x and ATM32xx-x1x devices*

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3.3 ATM33/e

- Monolithic Image (MSAP) for HID_Remote Application
- OTA MCUBoot Overwrite
- MSAP Bluetooth LE Feature Limitations
- HID_Remote Application Debugging Options
- Enable Fault Handling in Non-Secure Mode
- Do Not Use _printf_float for GCC nanolib
- Support External Flash For OTA Upgrade
- Sensor Hub Support
- Support for Bluetooth Privacy (Peripheral Mode) in HID_Remote Default Battery Type
- SPI Loopback Support
- Default battery type set to BATT_TYPE_NON_RECHARGEABLE

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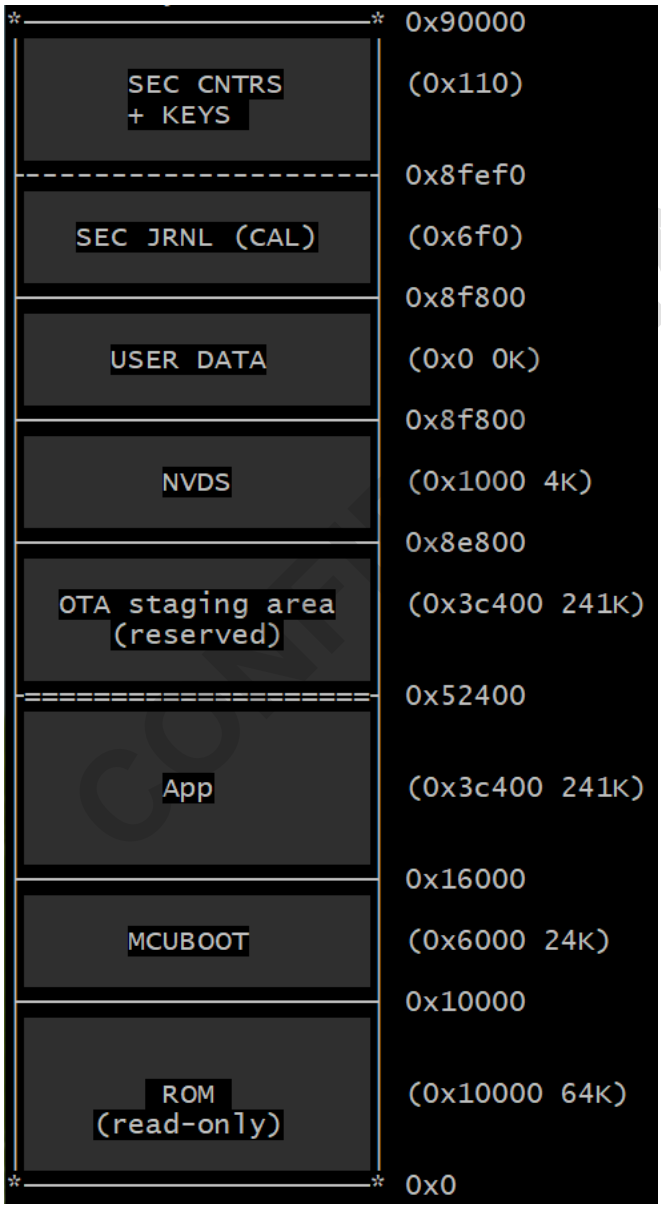
3.3.1 Monolithic Image (MSAP) for HID_Remote Application

This release allows the HID_Remote reference application to be built as a single Monolithic Stack & App Package (MSAP) that incorporates the Bluetooth LE host stack and controller. This replaces the use of the fixed non-upgradeable Bluetooth LE stack (ATMWSTK) image used in previous releases. The monolithic image uses a smaller stack library with a reduced feature set.

- Monolithic image must use an optimized Bluetooth LE Stack library called ATM MSAP and a new environment option `ATMWSTKLIB=PD50` for this purpose. An optional ATM_NO_SPE*=1 can further increase available space for the application by folding in some SPE functionality into the NSPE. This option however disables the ability to use Trustzone and implement isolated execution environments for security.
- Maximum image size for RCU application is 241 KB (using the MCUBoot overwrite option, and ATM_NO_SPE=1 see below). Use the command `layout_info` target to view the NVM layout. See below:

```
make BOARD=ATMEVK_3330e_QN USE_MCUBOOT=1 ATMWSTKLIB=PD50
USE_MCUBOOT_OVERWRITE=1 ATM_NO_SPE=1 layout_info
```

***Note**: The `ATM_NO_SPE` option is only supported for the RCU application.*



If the image swap strategy is used USE_MCUBOOT_OVERWRITE is omitted and the layout will include the MCUBOOT scratch area.

3.3.2 OTA MCUBoot Overwrite

The SDK now provides both overwrite and swap upgrade strategies while performing OTA updates. The overwrite strategy is mutually exclusive with the swap strategy. Only one can be employed at a time for a given MCUBoot image. The existing ATM OTA protocol offers a test boot option using MCUBoot's default swap strategy. With the overwrite strategy, the ability to perform a test boot is no longer available. The OTA protocol provides capabilities to detect when the device does not offer a test boot option. After a first boot activation, the firmware will be overwritten by MCUBoot and the system will boot to the new image. The first boot state will no longer be indicated to the remote host as the image is automatically marked as a GOOD image. There is a new environment option USE_MCUBOOT_OVERWRITE=1 for this purpose.

3.3.3 MSAP Bluetooth LE Feature Limitations

The Monolithic Stack & App Package (MSAP) is based on an optimized Bluetooth LE PD100 ATMWSTK variant with the following difference in features in order to optimize the footprint for RCU applications

- No Periodic Advertising
- No AoA/AoD
- L2CAP channels only support the standard FIXED channels
    - Supports only fixed CIDs 0x4-0x6 (Signaling, GATT/ATT, SMP)
    - No Dynamic Channels (0x40-0x7F)
    - No LE-credit based channels, No Enhanced Attribute Protocol (EATT)
    - No custom L2CAP protocols (no large SDU support)
- Host Stack messaging layer removed in favor of direct API calls that can be optimized by the linker.

3.3.4 HID_remote Application Debugging Options

The PD50 library can be used in a monolithic image without MCUBoot for testing purposes (OTA will be disabled). This allows you to use a larger debug image in the following combinations:

- When `DEBUG:=1` , Application Debug, Bluetooth LE stack debug

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- When DEBUG:=1 BLE_DEBUG:=, Application Debug, Bluetooth LE stack non-debug
- When DEBUG:= , Application non-debug, Bluetooth LE stack non-debug

When USE_MCUBOOT:=1, the build system will always choose to link to the non-debug Bluetooth LE stack to minimize the size of the image. Please keep this in mind when testing with MCUBoot with OTA support.

3.3.5 Enable Fault Handling in Non-Secure Mode

The default handling of the HardFault exception is in Secure mode as the AIRCR.BFHFNMINS bit is set to 0, and some fault cases should be handled in Secure mode. The ENABLE_NS_FAULTS option can be used if a user wants to handle UsageFault and MemManageFault in Non-Secure mode. It enables the faults and prevents them from escalating to a HardFault. However, the user should also be aware that additional exception handling and traceback code will be added to the Non-Secure image, resulting in an increase in code size.

3.3.6 Do Not Use _printf_float for GCC nanolib

The GCC nanolib does not have the capability to print floating point values without using a specific linker option. However, this option would result in an increased code size. It is recommended to change the unit of the value and print it as an integer, if possible. If it is necessary to print the value in floating point format, please include the build option CFG_GCC_PRINT_FLOAT=1 in the make command.

3.3.7 Support External Flash For OTA Upgrade

This release supports external flash for hosting OTA upgrades, in addition to data storage that was already supported. Use of external flash for OTA image staging frees up the available RRAM memory for application use. Please refer to the **SDK User Guide** on how memory is partitioned using an RRAM+External flash partition scheme. Supported devices include: Quad-SPI Macronix and both Single-SPI and Quad-SPI GigaDevice flash devices.

3.3.8 Sensor Hub Support

Demonstrates how to use the sensor hub to poll connected sensors while the CPU is in hibernation. Refer to the ATM_shub example.

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3.3.9 Support for Bluetooth Privacy (Peripheral Mode) in HID_Remote

HID_remote (RCU) now supports Bluetooth LE privacy feature. Please see HID_remote Application section.

3.3.10 SPI Loopback Support

Demonstrates the use of SPI driver to establish a loopback communication channel. Refer to SPI_loopback_client and SPI_loopback_server examples.

3.3.11 Default Battery Type Set to BATT_TYPE_NON_RECHARGEABLE for All Examples

All ATM33/e examples use the default battery type BATT_TYPE_NON_RECHARGEABLE in order to align with ATM33/e EVKs.

If using Li-ion battery please refer to the following documents:

ATM33/e Support for External Energy Storage Devices

- see Enable VBATLI for ATM33 and Enable VBATLI for ATM33/e sections

ATM33/e Series Evaluation Kit User Guide

- see Reconfigure the EVK section

ATM33/e Series Reference Manual

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3.3.12 Reference Applications and Examples

HID_remote Application

HID_remote(RCU) can now support Bluetooth LE privacy feature in the peripheral In addition board type settings can be set Atmosic RCU Reference Design (BOARD=ATMRC_3330e_HSC or BOARD=ATMRC_3330e_LIION)

Note: *See the HID_Remote Example Application Note for details.*

Tag Reference Application

AOA_dir_finding_tag demonstrates periodic advertisement with CTE (Constant Tone Extension) for direction finding applications. It also supports button press to enter the SOC OFF mode.

HID_Keyboard Reference Design

Added new BOARD name ATMKBD_3330e_HSC of HID_keyboard example.

4. SDK Platform Working Directory

Following is a list of all Atmosic SDK platform working directories:

- For ATM2/ATM3 devices use the following directories based on the device revision being used:
  - platform/atm2/ATM22xx-x0x
  - platform/atm2/ATM22xx-x1x
  - platform/atm3/ATM32xx-x0x
  - platform/atm3/ATM32xx-x1x
- For ATM3330/ATM3330e devices use the directory:
  - platform/atm33/ATM33xx-5

4.1 Reference Applications and Examples

The reference examples and applications newly added or renamed with this release are:

- ATM2/ATM3:
  - Same as SDK 5.3.0 release
- ATM33/e:
  - AOA_dir_finding_locator
  - AOA_dir_finding_tag
  - ATM_shub (Sensor Hub)
  - I2C_master
  - I2C_slave
  - motion_sensor_demo
  - SPI_loopback_client
  - SPI_loopback_server

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- HID_Keyboard
- HID_Remote
- Tag Reference application

5. ATM33/e Tx Power Settings

Tx output power settings default to 0 dBm or +4 dBm for the supported examples and applications listed in the Reference Examples and Applications section. To enable Tx output power higher than +4 dBm, please submit a support request for additional details.

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6. Limitations and Known Issues

6.1 ATM33/e

Issue	External SPI Flash used in quad mode will not be placed in deep sleep when not in use. [S-2707]
Impact	This impacts active state power consumption ONLY when quad SPI mode is configured. If the application is fully awake but is not actively using the external flash device (i.e. interface is closed), the external flash will not be placed into a deep sleep state until the system enters retention or hibernation. In retention or hibernate system power states, the external flash device will always be placed in deep sleep with the expected power consumption for that state. This issue does not affect single SPI mode operation (QSPI_SINGLE_MODE=1) and the device will be correctly managed in that configuration.
Resolution	To be fixed in a future release

Issue	[I2C Slave] Issue observed in longevity test [P-2353]
Impact	Only in long-run tests involving repeated device reset and data exchange.
Resolution	To be fixed in a future release

Issue	External SPI Flash pin assignment conflict in HID remote application [P-2322]
Impact	This issue occurs when external flash OTA is enabled in HID remote application (MCUBOOT_SECONDARY_EXT_FLASH=1). It would cause an OTA error in flash erasing. The cause is QSPI D1 and default PDM data input using the same pin P21. This issue does not affect the remote application with normal OTA and non-OTA configurations.The pin mux configuration shall reserve P18-P23 for QSPI, if external flash OTA is enabled.

Resolution	ATM3330/ATM3330e:
	1) Add the following defines in pinmap overlay file (e.g. pinmap_ATMEVK_3330e_QN_overlay.h)
	`#define PIN_PDM1_CLK 28`
	`#define PIN_PDM1_IN 29`
	2) Add `TEST_GROUP=2` option in make command to use PDM1 instance
	ATM3325 QK does not support external flash
	Use ATM3325 LQK has a built-in stacked flash if additional NVM is required

Issue	Low power clock driver blocks hibernation [P-2337]
Impact	Hibernation might be blocked after low power clock (RC oscillator based) calibration.
Resolution	To be fixed in a future release

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7. Resolved Issues

Following limitations and known issues are resolved in SDK 5.4.0.1 release.

7.1 ATM2/ATM3

Issue	HID_remote enabled Bluetooth LE privacy cannot reconnect to iPhone after OTA [G-2000]
Impact	Minor. Cannot reconnect to the iPhone.
Resolution	Resolved

7.2 ATM33/e

Issue	Cannot set BD address through AT command [P-1923]
Impact	Minor. This applies only for cases where the BD address is not programmed in the manufacturing floor. Typically, in most manufacturing setups SWD access is available to program this field.
Resolution	Resolved

Issue	ASSERT seen on system reset after flashing [P-1961]
Impact	Minor. This issue is seen intermittently (once in about 1.5 to 2 hours of continuous reboot testing. This issue may be seen when resetting the board after downloading new binary to the board.
Resolution	Resolved

Issue	Mobile OTA APP does not support OTA MCUBoot overwrite method. [SW-2484]

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Impact	If the RCU application enables the OTA MCUBoot overwrite method, the user cannot use Mobile OTA APP to upgrade the FW image.
Resolution	Resolved

Issue	RCU applications using Monolithic image will have build errors when using the ARM compiler. [SW-2504]
Impact	There is no problem using IAR and GCC.
Resolution	Resolved

Issue	Secure connection feature with ATMWSTKLIB=PD50 will result in runtime assert. [P-2172]
Impact	Cannot turn on the secure connection feature with ATMWSTKLIB=PD50. The default setting of RCU is legacy pairing not secure connect pairing mode.
Resolution	Resolved

Issue	Need to support ATMWSTKLIB=PD50 option build_archive [P-2173]
Impact	Cannot use build_archive to generate ISP atm file with option ATMWSTKLIB=PD50.
Resolution	Resolved

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Revision History

Date	Version	Description
March 24, 2023	1.10	Updated ATM Log Level, corrected typos.
March 8, 2023	1.00	Initial version created.

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www.atmosic.com